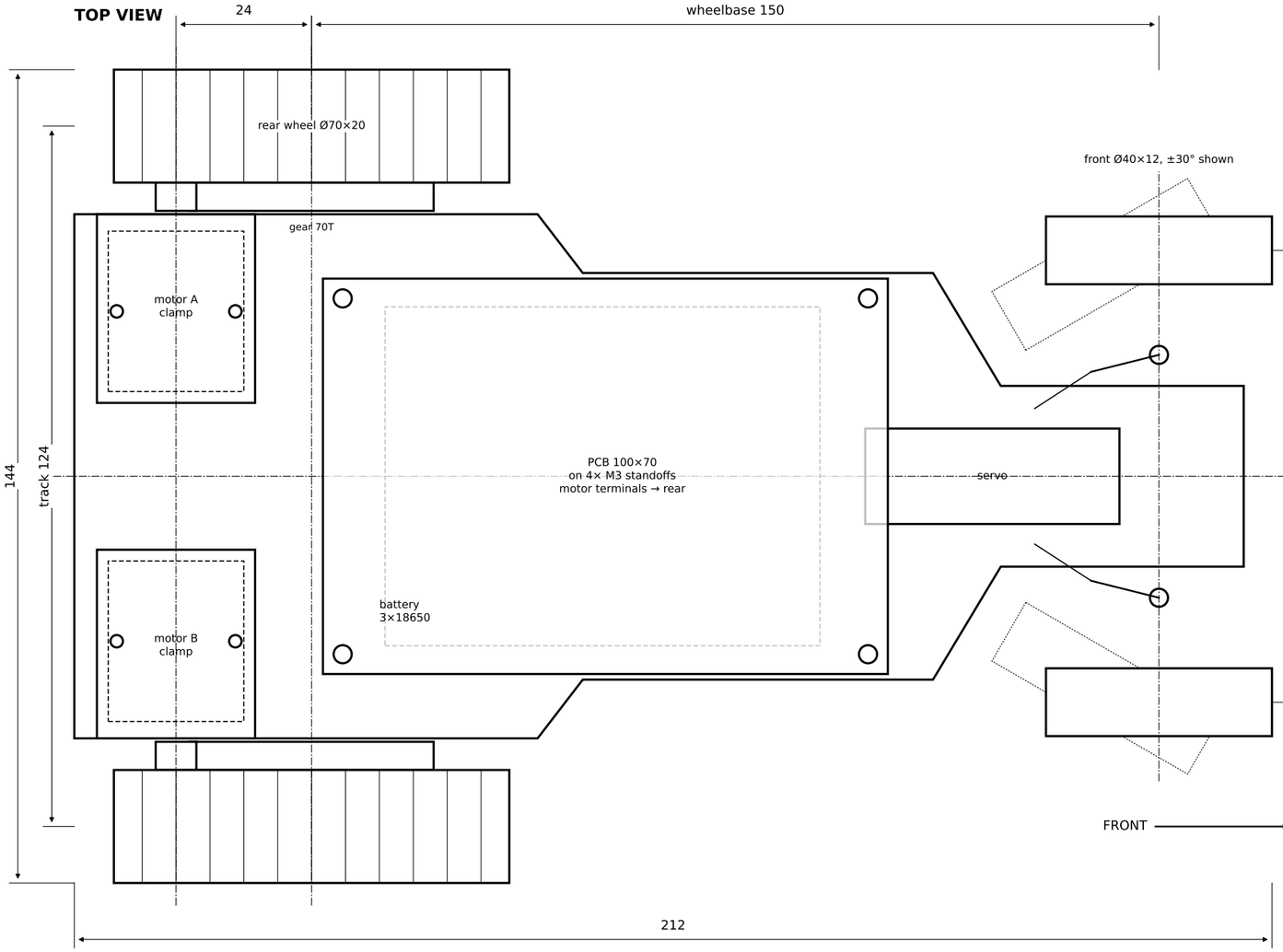
TOP VIEW



Title	**DIY-Auto — concept v2 (for approval)**
Scale	1:1 on A3
Units	mm
Projection	third angle
Date	2026-09-26
Drawn / checked	Claude / Ruben

DESIGN DECISIONS — for approval

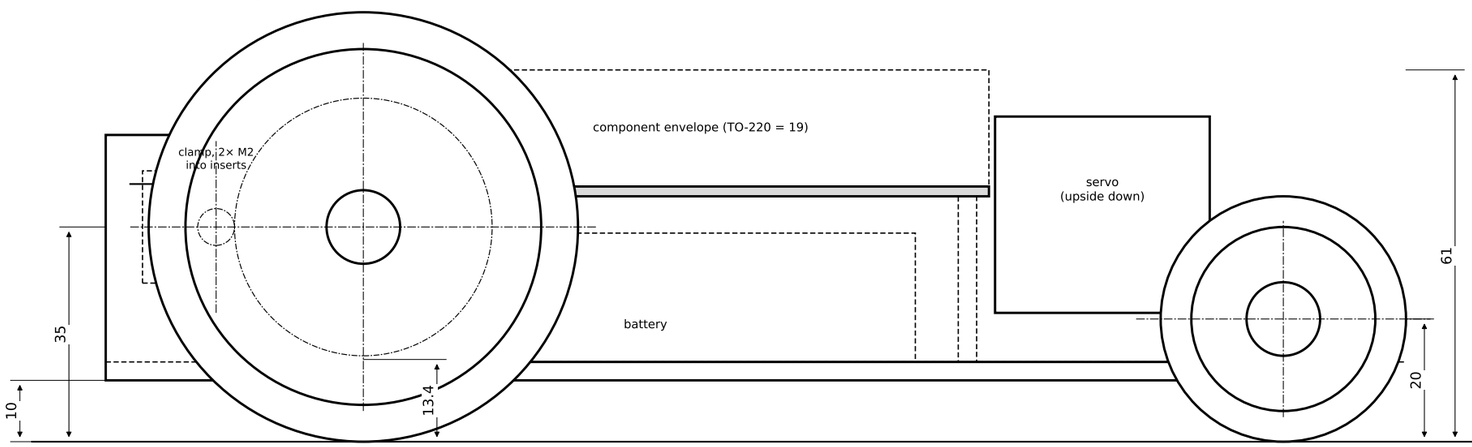
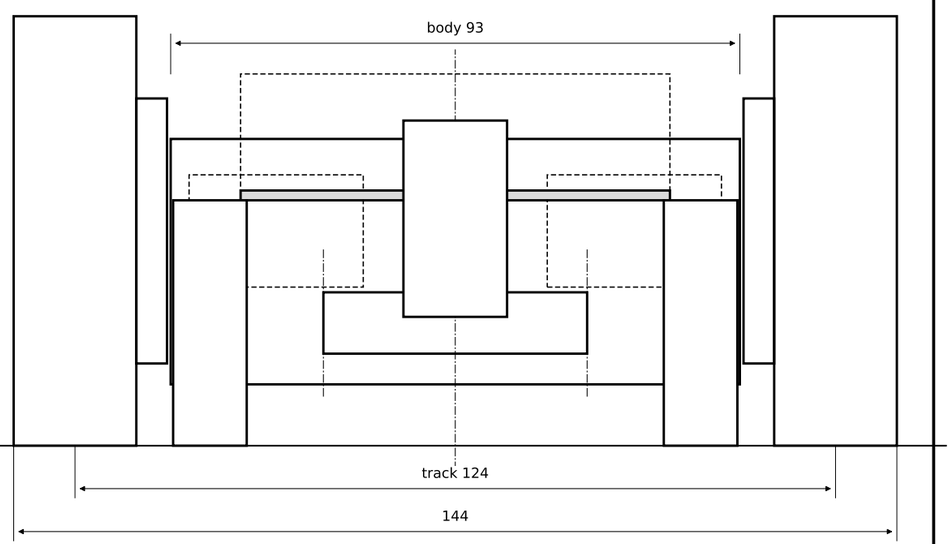
1 Rear wheel drive, one motor per wheel. Front: C-hub knuckles on M3 kingpins, ±30°, servo GWS Mini STD upside down, horn at arm height, Ackermann arms. TPU tyres front and rear.
2 Single stage 7:1: motor pinion 10T → 70T on the inside of the rear wheel, module 0.6, gear Ø42 (tip Ø43.2), centre distance 24, width 5.
3 Axles do not turn: Ø5 steel fixed in body or knuckle, each wheel on 2× MR105 in its hub. 8 bearings in total (you have 10).
4 Motor slides into a cradle in the body (the flats stop it rotating), a clamp on top holds it with 2× M2 screws into M2 inserts.
5 Motors behind the rear axle, battery in front of it: centre of gravity ≈ 40 mm in front of the rear axle → ≈ 74 % on the driven wheels.
6 PCB gets a buck converter 12 V → 5 V (servo + ESP32) and a 3-pin servo header. The powerbank goes.

TO CHECK BEFORE MODELLING

a Motor pinion tip diameter: 7.2 mm confirms module 0.6.
b Battery holder: 77 × 60 × 21 assumed (3 cells side by side).
c Servo: length over the mounting tabs (45 assumed) and the horn height.
d Speed: ≈ 6.3 m/s if the motor runs ≈ 12 000 rpm under load (estimate).

Pos	Part	Qty	Material
1	Body (floor, walls, motor cradles, servo pocket)	1	PETG
2	Motor clamp	2	PETG
3	Rear rim with 70T m0.6 gear, 2 bearing seats	2	PETG, 0.2 nozzle
4	Front rim Ø30, 2 bearing seats	2	PETG
5	Tyre rear Ø70×20 / front Ø40×12	2+2	TPU 95A
6	C-hub knuckle / tie rod Ø1.5 wire / M3 kingpin + nylock	2/2/2	PETG / steel
7	Axle Ø5 (4 short stubs)	4	steel
8	MR105-2RS 5×10×4	8	have (10)
9	Shaft collar Ø5	4	buy
10	M2 threaded insert + M2 screw	4+4	buy
11	Servo GWS Mini STD 35×16.9×32	1	have
12	Motor 260-size, pinion 10T m0.6	2	have
13	Battery holder 3×18650	1	have
14	PCB DIY-Auto (+ buck converter, servo header)	1	JLCPCB

SIDE VIEW (left)



FRONT VIEW